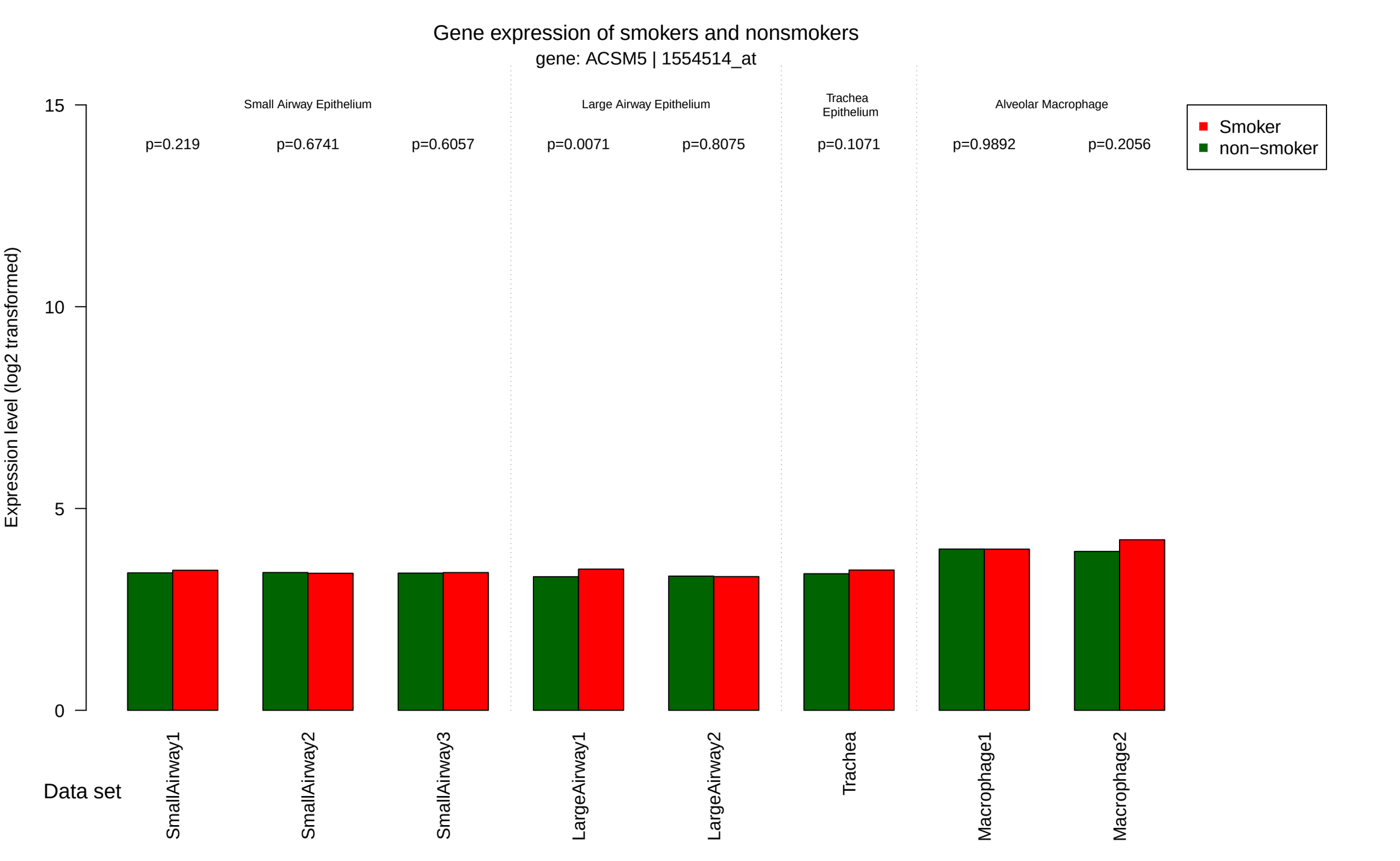
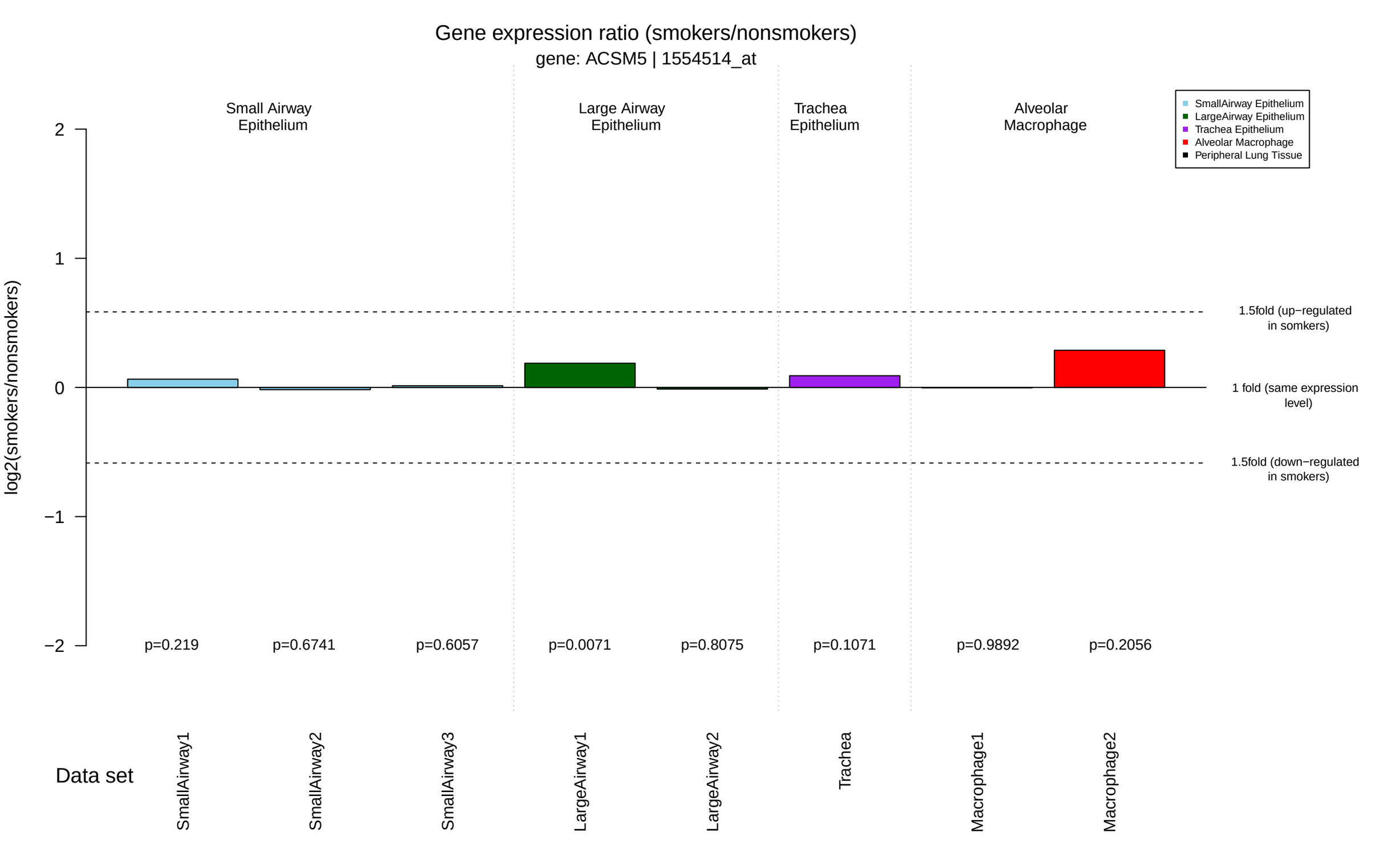
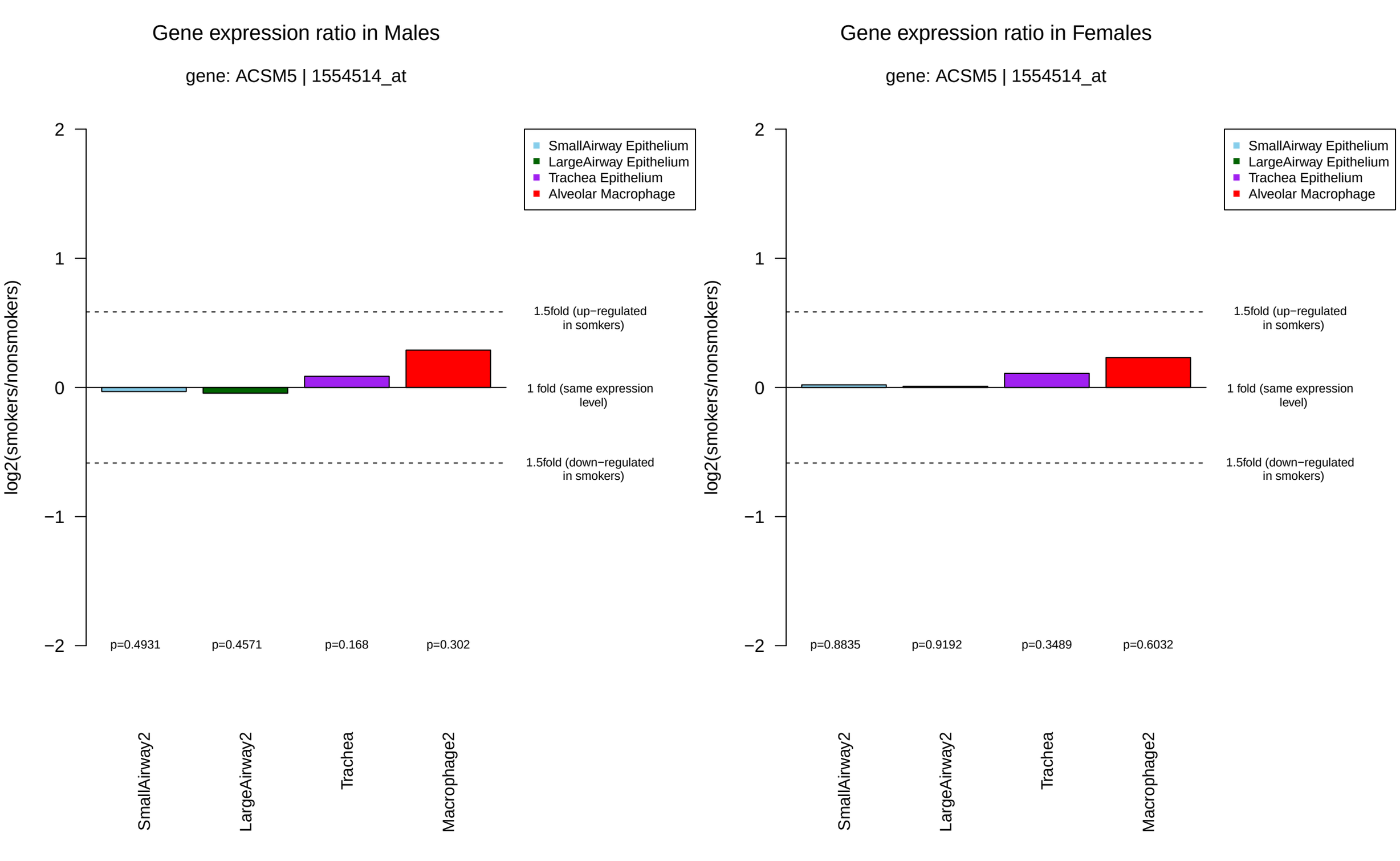
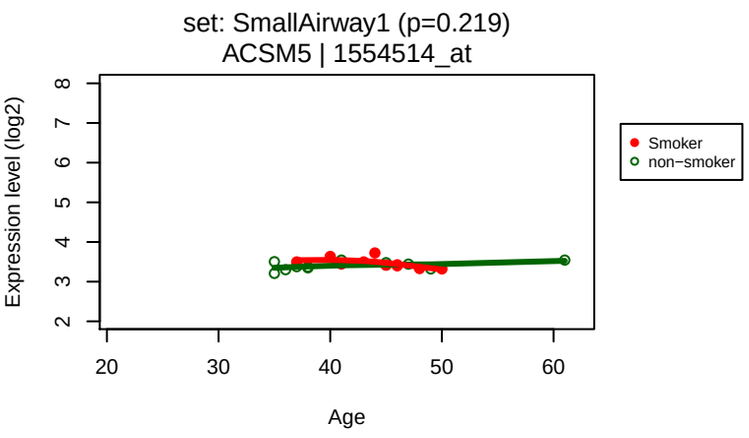
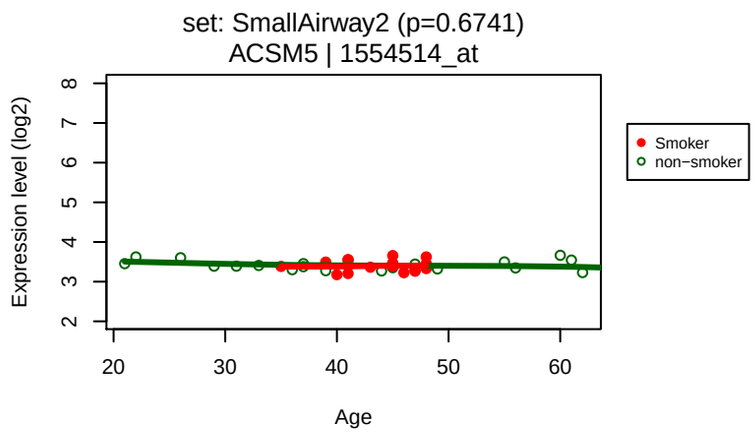
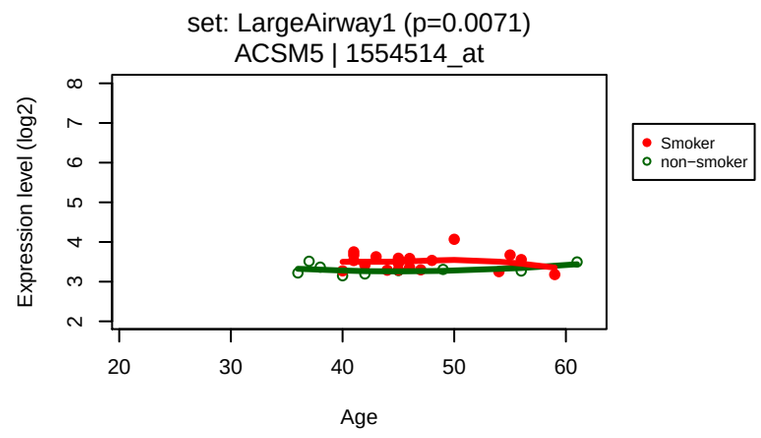

Gene expression of smokers and nonsmokers
gene: ACSM5 | 1554514_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.219
p=0.6741
p=0.6057
p=0.0071
p=0.8075
p=0.1071
p=0.9892
p=0.2056
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

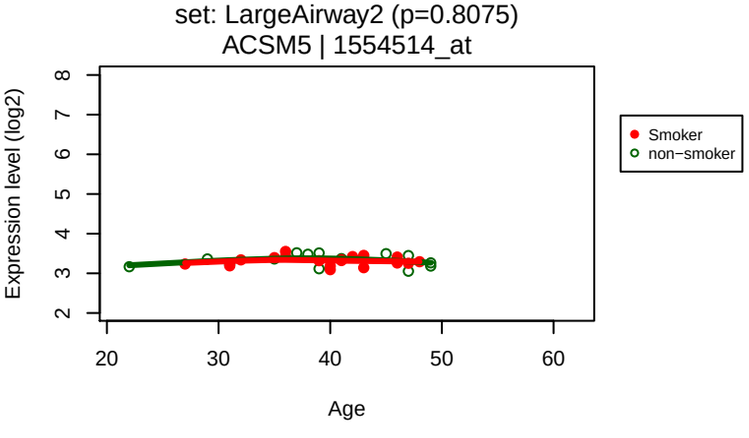

Gene expression ratio (smokers/nonsmokers)
gene: ACSM5 | 1554514_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.219
p=0.6741
p=0.6057
p=0.0071
p=0.8075
p=0.1071
p=0.9892
p=0.2056
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

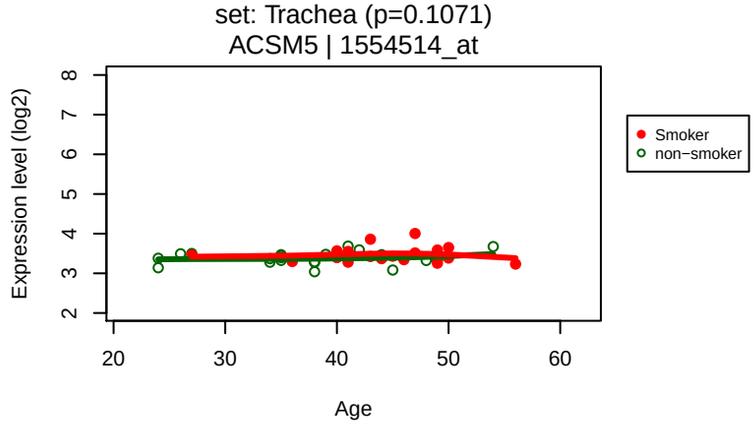

Gene expression ratio in Males
gene: ACSM5 | 1554514_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4931
p=0.4571
p=0.168
p=0.302
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSM5 | 1554514_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8835
p=0.9192
p=0.3489
p=0.6032
SmallAirway2
LargeAirway2
Trachea
Macrophage2

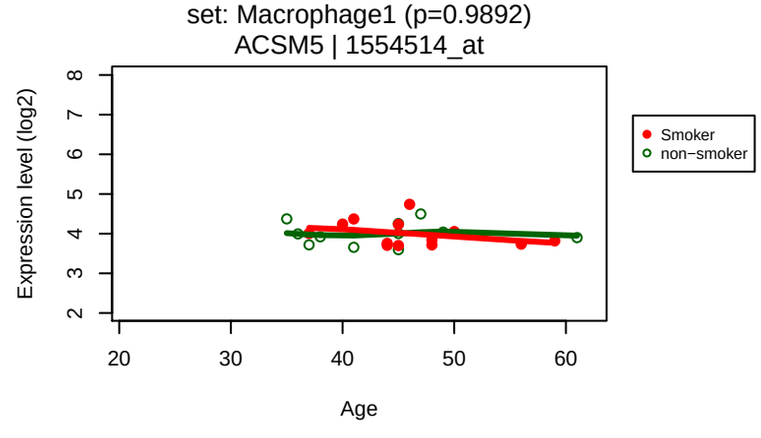

set: SmallAirway1 (p=0.219)
ACSM5 | 1554514_at
Expression level (log2)
Age
Smoker
non−smoker

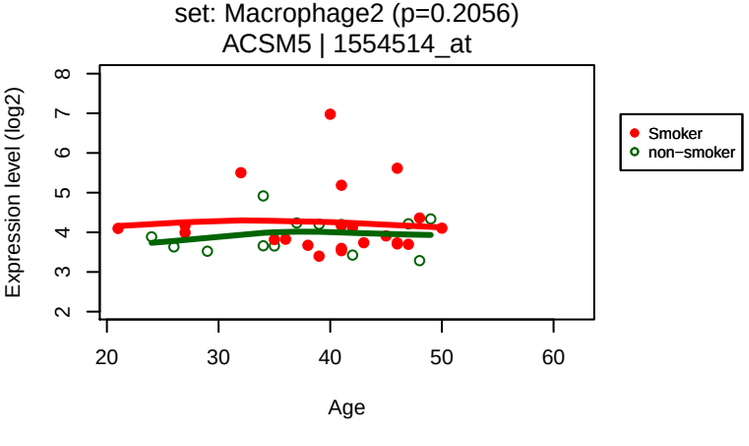

set: SmallAirway2 (p=0.6741)
ACSM5 | 1554514_at
Expression level (log2)
Age
Smoker
non−smoker

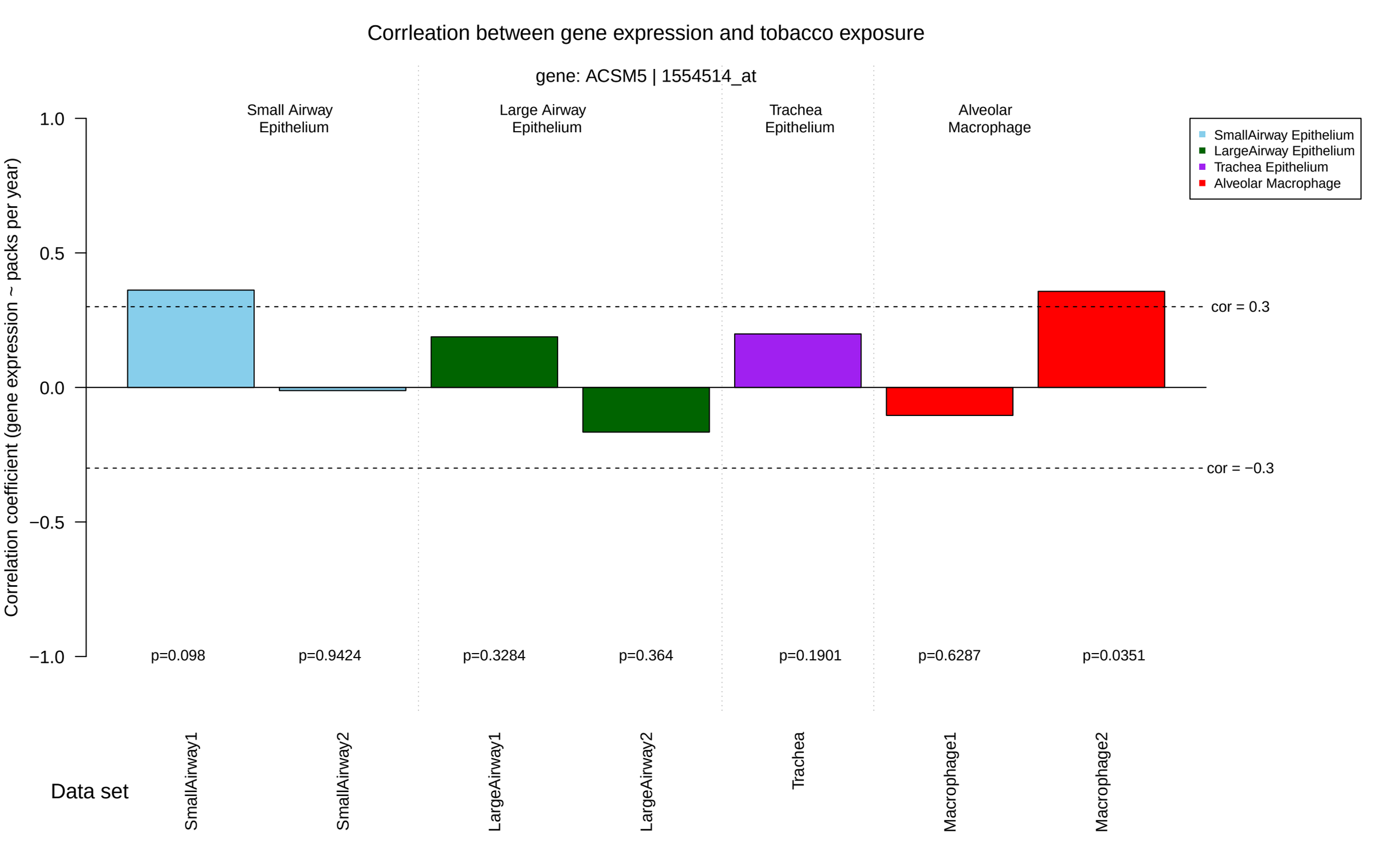

set: LargeAirway1 (p=0.0071)
ACSM5 | 1554514_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8075)
ACSM5 | 1554514_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1071)
ACSM5 | 1554514_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9892)
ACSM5 | 1554514_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2056)
ACSM5 | 1554514_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSM5 | 1554514_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.098
p=0.9424
p=0.3284
p=0.364
p=0.1901
p=0.6287
p=0.0351
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2